		$\therefore v_P = \sqrt{2}v_Q = \sqrt{2}v$
11	**B**	$\phi = -\frac{GM}{r} \Rightarrow \phi \propto \frac{M}{r}$ $\frac{\phi_M}{\phi_E} = \frac{M_M}{M_E} \times \frac{r_E}{r_M} \Rightarrow \frac{\phi_M}{-63} = \frac{0.012M_E}{M_E} \times \frac{D_E/2}{0.25D_E/2}$ $\phi_M = \frac{0.012}{0.25} \times (-63) = -3\ MJ\ kg^{-1}$
12	**B**	Particle in simple harmonic motion $a_0 = \omega^2 x_0$